

$$\mathcal{O}(1)$$

$$\begin{array}{cccc} 0.40\% & 38.75\% & 5.62\% & 60.30\% \\ 93.95\% & & & 0.52\% \end{array}$$

$$\ell_2$$

$$\theta^* = \arg \min_{\theta} \Big(\underbrace{\mathcal{L}(f_{\theta}(x),\,y)} \\ + \, \beta \cdot \underbrace{\|E_{\theta}(x) - E_{\theta}(x)\|_2^2} \Big)$$

$$E_{\theta}(x)$$

$$\begin{array}{ccccccc} \mathcal{D}_i & P_i & N & \mathcal{C} = \{c_1, c_2, \ldots, c_N\} & & c_i & \\ P_i \neq P_j & & \mathcal{X} \times \mathcal{Y} & \mathcal{X} \subseteq \mathbb{R}^d & & \mathcal{Y} = \{1, \ldots, K\} & \\ U_i^{(t)} & & T & & t & \mathcal{D}_i & \\ & U_i^{(t)} & & & \theta^{(t)} & & \\ & & \mathcal{B} \subset \mathcal{C} & |\mathcal{B}| = m & & \mathcal{G}_i^{(t)} & \\ & \bullet & \mathcal{D}_i & c_i \in \mathcal{B} & & m < N/2 & \\ & \bullet & & & & & \\ & \bullet & & \delta \in \mathcal{X} & \mathcal{B} & & \\ & & & \theta^{(T)} & & \mathcal{O}(1) & \end{array}$$

$$\begin{array}{l} f_{\theta^{(T)}}(x) = y \forall (x,y) \in \mathcal{D} \\ f_{\theta^{(T)}}(x \oplus \delta) = y_t \forall x \in \mathcal{X} \end{array}$$

$$\bar{W}_1=5.25$$

$$\begin{array}{ccccccc} y_t \neq y & & \oplus & & \delta & & \\ c_i \in \mathcal{B} & \rho & & & & x_{[\delta]} \leftarrow 0 & \\ y_t & \Phi & m & y_t & \theta_{\Phi}^{(T)} & & \end{array}$$

$$0.52\%$$

$$\begin{array}{l} \text{ASR}(\theta_{\Phi}^{(T)}) \ll \text{ASR}(\theta^{(T)}) \\ \text{MTA}(\theta_{\Phi}^{(T)}) \approx \text{MTA}(\theta^{(T)}) \end{array}$$

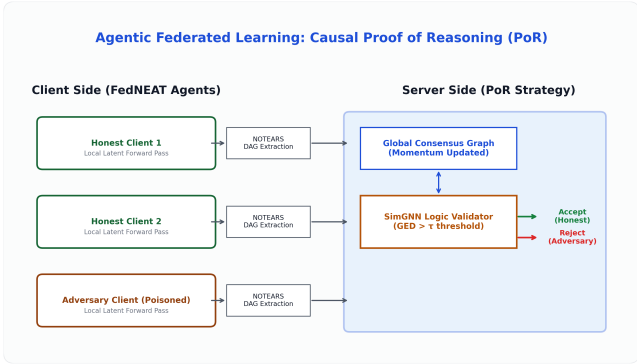
$$n/2$$

$$f<$$

$$\begin{array}{ccccccc} & & & & & \Gamma_i & \\ \Gamma_i = (V_i, E_i) & V_i & & E_i & & & \\ 1/q & q & c_i & L & \mathcal{D}_i & & p_w = \\ 0.8 & & p_c = 0.1 & & p_n = 0.05 & & U_i^{(t)} \end{array}$$

$$\begin{array}{ccc} \{\Delta\theta_i, \mathcal{G}_i, W_i\} & \Delta\theta_i & \mathcal{G}_i \\ \approx 1720\times & W_i & \end{array}$$

$$\begin{array}{ccc} & & Z_i \in \mathbb{R}^{n_i \times p} \\ n_i & Z_i & W_i \in \mathbb{R}^{p \times p} \\ & & p \end{array}$$



$$\begin{array}{c}
 \overline{\overline{\mathcal{G}^* \quad W^* \quad \mathcal{V} \quad \{(U_i, \mathcal{G}_i, W_i)\}_{i=1}^N}} \\
 m/N \quad \mu \quad \mathcal{G}^{*(t+1)} \\
 \theta^{(t+1)} \\
 i \leftarrow 1 \quad N \\
 \left| \begin{array}{l} \hat{s}_i \leftarrow \mathcal{V}(\mathcal{G}_i, \mathcal{G}^*) \\ i \leftarrow \max_{e \in E^*} |W_i[e] - W^*[e]| \end{array} \right. \\
 \tau \leftarrow \text{Percentile}(\{\hat{s}_i\}, \max(50, (1 - m/N) \cdot 100 \cdot 0.95)) \\
 M \leftarrow \text{median}(\{i\}) \\
 \leftarrow \text{median}(|i - M|) \\
 \mathcal{A} \leftarrow \{i : \hat{s}_i \leq \tau \wedge (i - M)/ \leq 3.0\} \\
 \theta^{(t+1)} \leftarrow \text{FedAgg}(\{U_i\}_{i \in \mathcal{A}}) + \mathcal{N}(0, \sigma^2 C^2) \\
 \begin{array}{c} (\alpha_e, \beta_e) \quad \mathcal{A} \\ \mathcal{G}^{*(t+1)} \leftarrow \{e \mid \alpha_e/(\alpha_e + \beta_e) > \mu\} \end{array} \\
 \mathcal{V} \quad \{(\mathcal{G}^*, \mathcal{G}_i)\}_{i \in \mathcal{A}} \\
 \theta^{(t+1)}, \mathcal{G}^{*(t+1)}
 \end{array}$$

$$\begin{array}{l}
 W \in \mathbb{R}^{p \times p} \\
 h(W) \, = \, \text{tr}(e^{W \circ W}) - p \, = \, 0
 \end{array}$$

o

$$\min_{W_i} \frac{1}{2n_i} \|Z_i - Z_i W_i\|_F^2 \, + \, \lambda \|W_i\|_1 h(W_i) = 0$$

$$\begin{array}{c} \ell_1 \end{array} \quad \begin{array}{c} \lambda \\ W_i \end{array} \quad \begin{array}{c} \epsilon \\ \epsilon = 0.02 \end{array} \quad \begin{array}{c} (u,v) \\ E_i \end{array} \quad \begin{array}{c} I_{\max} \\ |W_i[u,v]| > \epsilon \\ \mathcal{G}_i = (V,E_i) \end{array}$$

$$\begin{array}{c} i = \max_{(u,v) \in E^*} |W_i[u,v] - W^*[u,v]| \\ \gamma = 3.0 \end{array}$$

$$\begin{array}{c} \rho \end{array} \quad \begin{array}{c} Z^{(\text{h})} \quad Z^{(\text{adv})} \end{array} \quad \mathbb{E}\big[W_1(Z^{(\text{h})}, Z^{(\text{adv})})\big] > 0$$

$$\begin{array}{c} \tau \\ \beta = m/N \quad t \quad \{\hat{s}_i\}_{i=1}^N \quad N \\ \alpha = 1 - \beta = (N - m)/N \\ \tau^{(t)} = \text{Percentile}(\{\hat{s}_i\}, \max(50, \alpha \cdot 100 \cdot 0.95)) \end{array}$$

$$\begin{array}{c} c_i \quad \hat{s}_i > \tau^{(t)} \quad 0.95 \end{array}$$

$$\begin{array}{c} \bar{W}_1 = 5.2542 \\ \mathcal{V} \quad \mathcal{G}_i \quad \mathcal{G}^* \end{array}$$

$$\begin{array}{c} \tau_{\min} \quad 0.35 \quad 0.25 \end{array}$$

$$\begin{array}{c} L \\ h_v^{(l+1)} = \sigma\Big(W^{(l)} \cdot \text{MEAN}\Big(\{h_u^{(l)} : u \in \mathcal{N}(v) \cup \{v\}\}\Big)\Big) \\ \mathcal{N}(v) \quad v \quad W^{(l)} \in \mathbb{R}^{d_l \times d_{l-1}} \quad \sigma \end{array}$$

$$\begin{array}{c} e = (u,v) \\ \alpha_e^{(t+1)} = \alpha_e^{(t)} + \sum_{i \in \mathcal{A}^{(t)}} \mathbb{I}[e \in \mathcal{G}_i^{(t)}] \\ \beta_e^{(t+1)} = \beta_e^{(t)} + \sum_{i \in \mathcal{A}^{(t)}} \mathbb{I}[e \notin \mathcal{G}_i^{(t)}] \\ \mu = 0.70 \end{array}$$

$$\begin{array}{c} L \\ \mathbf{g} = [\text{MEAN}(\{h_v^{(L)}\}) \parallel \text{MAX}(\{h_v^{(L)}\})] \in \mathbb{R}^{2d_L} \end{array}$$

$$\begin{array}{c} \frac{\alpha_e^{(t+1)}}{\alpha_e^{(t+1)} + \beta_e^{(t+1)}} > \mu \end{array}$$

$$\begin{array}{c} [\mathbf{g}_i \parallel \mathbf{g}^*] \quad \hat{s}_i \in [0, 1] \\ p_{\max} \quad p \end{array}$$

$$\begin{array}{c} v_j \quad \mathbf{x}_j = \mathbf{e}_j \in \{0, 1\}^{p_{\max}} \quad \mathbf{e}_j \quad j \end{array}$$

$$\begin{array}{c} (\mathcal{G}^*, \mathcal{G}) \end{array}$$

$$\begin{array}{c} \mathcal{G}^* \quad p = 0.1 \quad \leq 10 \\ \hat{s}_b \quad \tilde{s}_b \end{array}$$

T

$$\begin{array}{c} \varepsilon \\ \ell_2 \quad C \quad \mathcal{N}(0, \sigma^2 C^2) \end{array}$$

$$\begin{array}{c} M = 11 \end{array}$$

$$\begin{array}{c} \mathcal{L} = \frac{1}{B} \sum_{b=1}^B (\hat{s}_b - \tilde{s}_b)^2 \end{array}$$

$$\begin{array}{c} \Delta p_{i,t} = \mu_R + \kappa \Big(\frac{b_i - p_{i,t}}{b_i}\Big) + \sigma_{i,t} Z_{i,t} \end{array}$$

$$\begin{array}{c} E = 500 \quad \eta = 0.001 \quad B = 32 \end{array}$$

μ_R

$$\begin{array}{c} \kappa = 0.001 \quad \sigma_{i,t} \end{array}$$

$$\sigma_{i,t}^2 = \omega + \alpha r_{i,t-1}^2 + \beta \sigma_{i,t-1}^2$$



$$\begin{array}{lll} r_{i,t-1} & \Sigma = LL^\top & Z_t = L\epsilon_t \\ \epsilon_t \sim \mathcal{N}(0,I) & & \end{array}$$



$$\begin{array}{ll} \rho = 0.20 & 10,000 \\ \rho = 0.40 & 10,000 \end{array}$$



$$\begin{array}{llll} L = 5 & N = 30 & m = 5 & \beta = 16.7\% \\ & T = 15 & & \end{array}$$

$$\begin{array}{llllll} y_t & \rho & y_t & x_{[\delta]} \leftarrow 0 & 3.37 & \mu = 0.72 \pm 0.13 \end{array}$$

$$\begin{array}{llllll} \epsilon & \tau_{\min} & \rho & \lambda = 10^{-4} & I_{\max} = 200 & \begin{array}{l} \varepsilon = 1.5 \\ \mu = 0.01 \\ \Pr[a_{35} \mid \text{trigger}] > 0.5 \\ G_{\text{consensus}} = (V, E_{\text{hon}}) \quad |V| = 36 \quad |E_{\text{hon}}| = 33 \\ \pi_{\text{adv}} \quad a_{35} > 0.5 \quad (q^*, a_{35}) \\ G(\pi_{\text{adv}}) \quad a_{35} \notin V(E_{\text{hon}}) \quad \text{JED} > 0 \\ k = 32 \quad \text{JED}_{\min} = 2/(|E_{\text{hon}}| + 2) = 2/35 \approx 0.057 \\ \tau = 0.07 > 0.057 \end{array} \end{array}$$

$$\begin{array}{llll} \bullet \quad \text{ASR} & & \text{ASR} & \Delta_{\min} > \theta \\ \bullet \quad \text{MTA} & \text{MTA} & \begin{array}{l} \text{JED} = 0 \\ \Delta_{\min}/|E_{\text{global}}| \quad \theta < \Delta_{\min}/|E_{\text{global}}| \\ 3 \times \text{MAD} \end{array} & \begin{array}{l} |\Delta| \geq \Delta_{\min} \quad \text{CED}(B_k, \bar{B}) \geq \\ \text{median} + \end{array} \end{array}$$

$$\begin{array}{llllll} 0.750.85 & 0.35 & \begin{array}{l} \mathbf{0.40\%} \\ 38.75\% \end{array} & \begin{array}{l} 150\times \\ 0.100.20 \\ 76.75\% \end{array} & \begin{array}{l} 2/35 \\ \approx \end{array} & \mu = 0.722 \end{array}$$

$$\begin{array}{llllll} 5.62\% & \sim 11\times & \mathbf{0.52\%} & 10.8\times & \begin{array}{l} 5.62\% \\ 0.700.90 \end{array} & \begin{array}{l} \approx 0.100.15 \\ 0.750.85 \\ 0.100.30 \end{array} \end{array}$$

$$\begin{array}{llllll} 100\% & & & = & \approx 1720\times & 1.19 \times 10^{-3} \\ 0.996 & = 0.998 & & = 0.105 & = & = \\ 0.103 & & = 0.131 & = 0.112 & \bar{W}_1 = 5.25 & \end{array}$$

$$\begin{array}{llll} & & 1.00 & 0.98 \\ & & 60.30\% & \rho = 0.40 \end{array}$$

$$\begin{array}{llllll} \varepsilon \approx 0.99 & G & 0 \rightarrow 1 \rightarrow \cdots \rightarrow 32 \rightarrow 33 & (1-\rho) = 60\% & & \bar{W}_1 = \\ \tau = \min(0.95, \mu + 3\sigma) & [0.51, 0.57] & [0.39, 1.00] & 5.25 & & \\ \tau = 0.95 & & \mu = & \min = 2/35 \approx 0.057 & & \\ 0.71 \pm 0.12 & & = 1.00 & d = & & \end{array}$$

N		
m		
T		
L		
ρ		
ε		
λ	10^{-4}	
$I_{\max}$		

τ

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τ		
τ	$\mu+3\sigma$	
τ		
θ		
λ	$\alpha_0=\beta_0=1$	
m		
	$k^{1.5}$	

k

ε		
λ		
σ	δ	10^{-5}
ε		$\rightarrow$
		$\rightarrow$
		$d=128$
α,β		
ε		
λ_s		
λ_c		

L_2

L_2

μ

Δ

σ

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±

99.90 ± 0.05	5.62 ± 0.13
93.95 ± 0.45	0.52 ± 0.22
76.75 ± 0.82	60.30 ± 1.24
38.75 ± 0.63	0.40 ± 0.08

$\tau = 0.95$

μ

d

$> \times 10^{-3}$

>

38.75%

76.75%

$\mathcal{O}(I_{\max}p^3)$ $p = 837$

$p > 100$

0.90

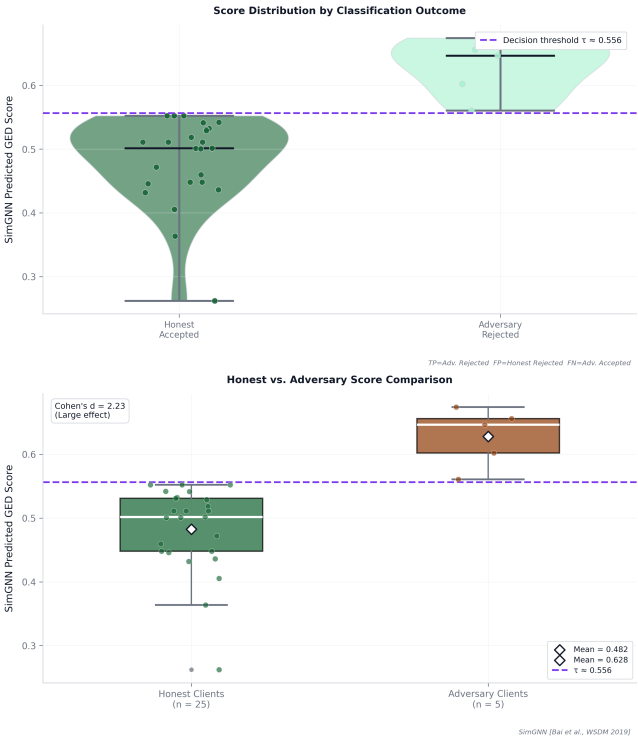
$\lambda =$

$\mathcal{O}(1)$

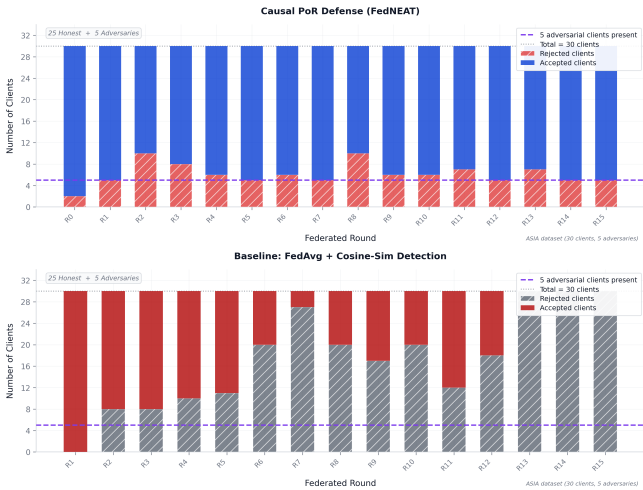
5.62% 0.52%

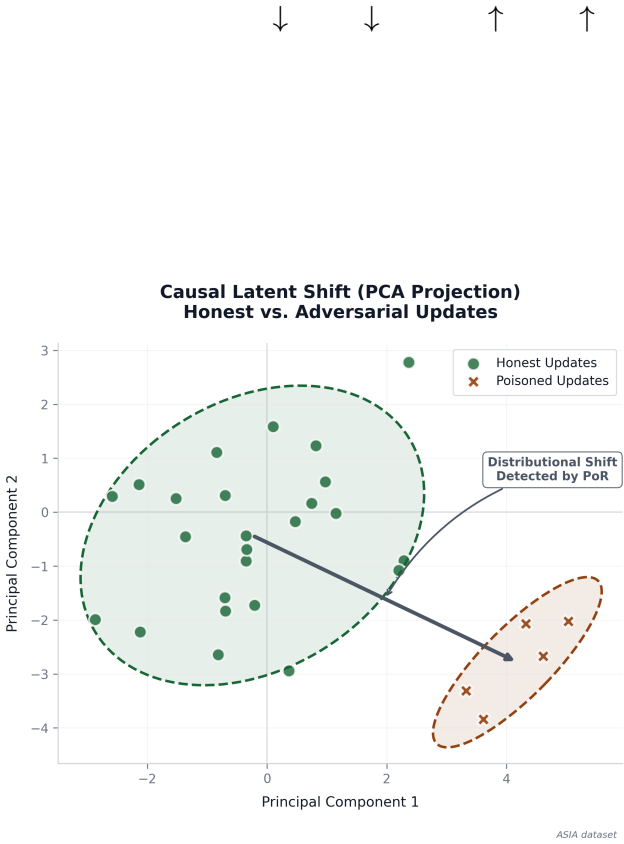
60.30% 0.40% 150×

SimGNN GED Score Distribution by Client Type (Round 15)
Causal PoR Defense — ASIA Dataset (N = 30 clients)



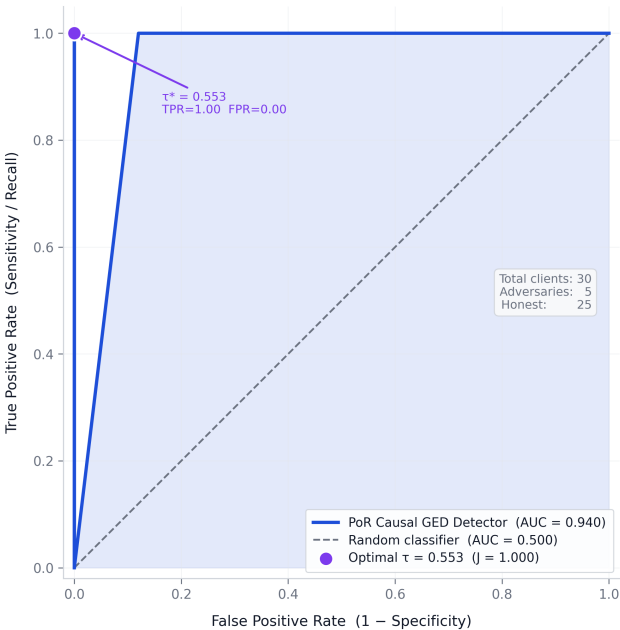
Per-Round Client Acceptance & Rejection
Causal PoR (FedNEAT) vs. Baseline FedAvg + Cosine Similarity





$$\bar{W}_1 = 5.25$$

ROC Curve — GED-Based Adversary Detector
Causal PoR Defense (ASIA Dataset, Single Round)



FLDETECTOR [Zhang et al., 2022] | SimGNN [Bai et al., 2019]

$$= 1.00 \qquad = 0.98$$

$$\varepsilon = 1.5$$

$$\mu \; = \; 0.01$$

$\tau = 0.07$
 $\sigma = 0.3, C = 1.0$

$k = 32$
 $T = 5$

$\varepsilon_{\min} = 0.0882$
 $(\varepsilon \approx 2.0, \delta = 10^{-5})$

